Addition-Elimination Reactions of Acyl Chlorides (Cambridge (CIE) A Level Chemistry): Revision Note

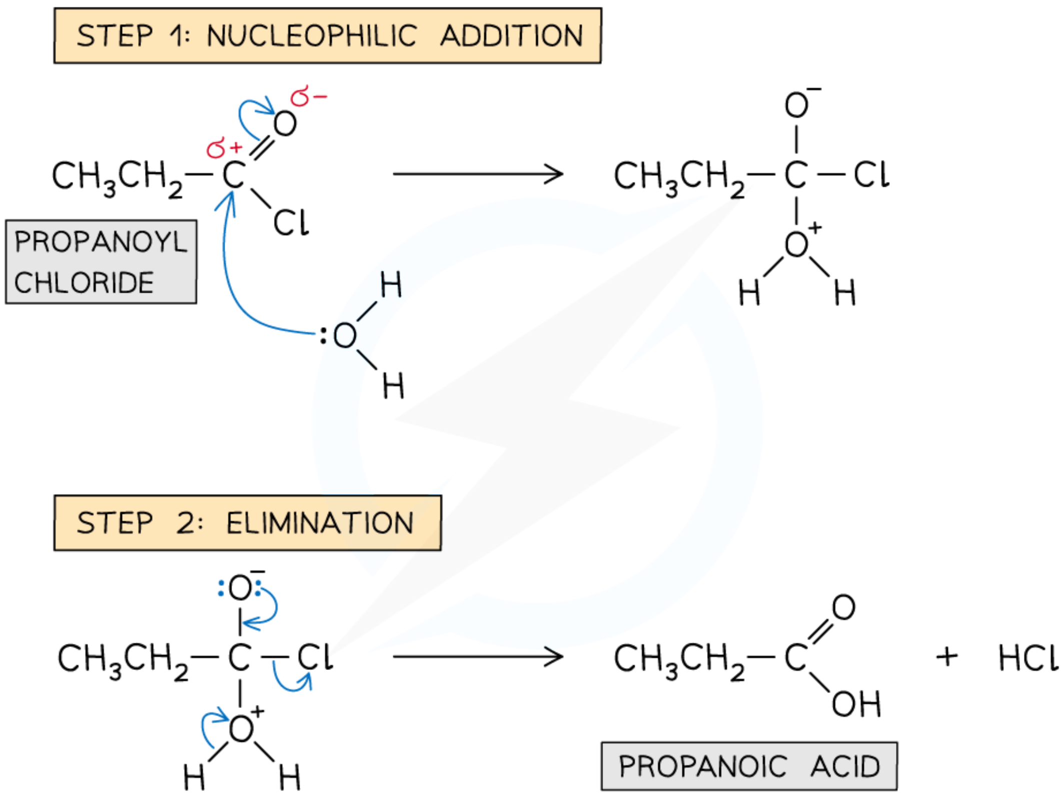
Mechanism of Addition - Elimination in Acyl Chloride Reactions

- Acyl chlorides undergo **addition-elimination** reactions such as **hydrolysis, esterification** reactions to form esters, and **condensation** reactions to form **amides**
- The general mechanism of these addition-elimination reactions involves two steps:
  - **Step 1 -** Addition of a **nucleophile** across the C=O bond
  - **Step 2 -** Elimination of a **small molecule** such as HCl or H_2O

Mechanism of hydrolysis of acyl chlorides

- In the **hydrolysis** of acyl chlorides, the water molecule acts as a **nucleophile**
  - The lone pair of the oxygen atom from water carries out an **initial attack** on the carbonyl carbon
  - This is followed by the elimination of a hydrochloric acid (HCl) molecule

Reaction mechanism of the hydrolysis of acyl chlorides



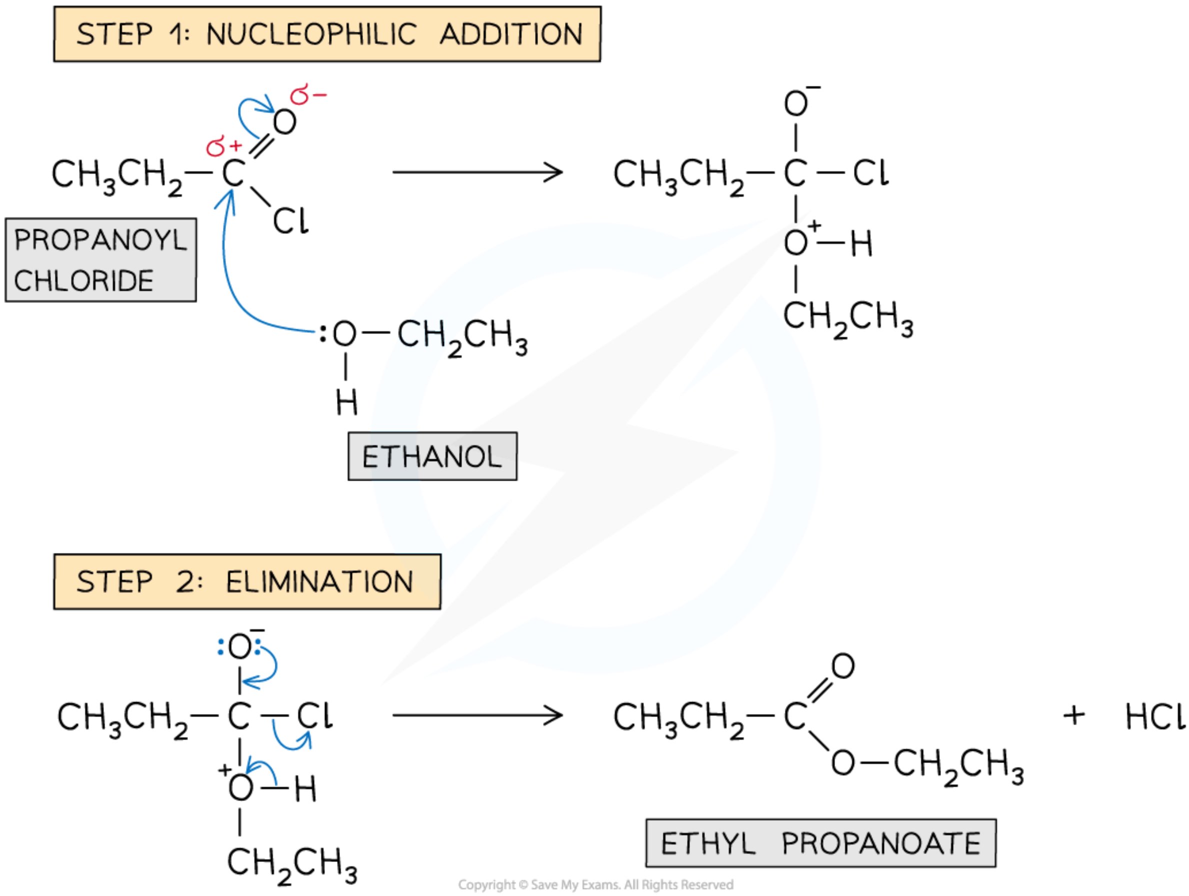
The two-step addition-elimination reaction mechanism of propanoyl chloride to form propanoic acid

Formation of esters: reaction mechanism

- In the **esterification** reaction of acyl chlorides, the alcohols or phenols act as a **nucleophile**
  - The lone pair of the alcohol / phenol oxygen atom carries out an **initial** attack on the carbonyl carbon
  - This is again followed by the elimination of an HCl molecule

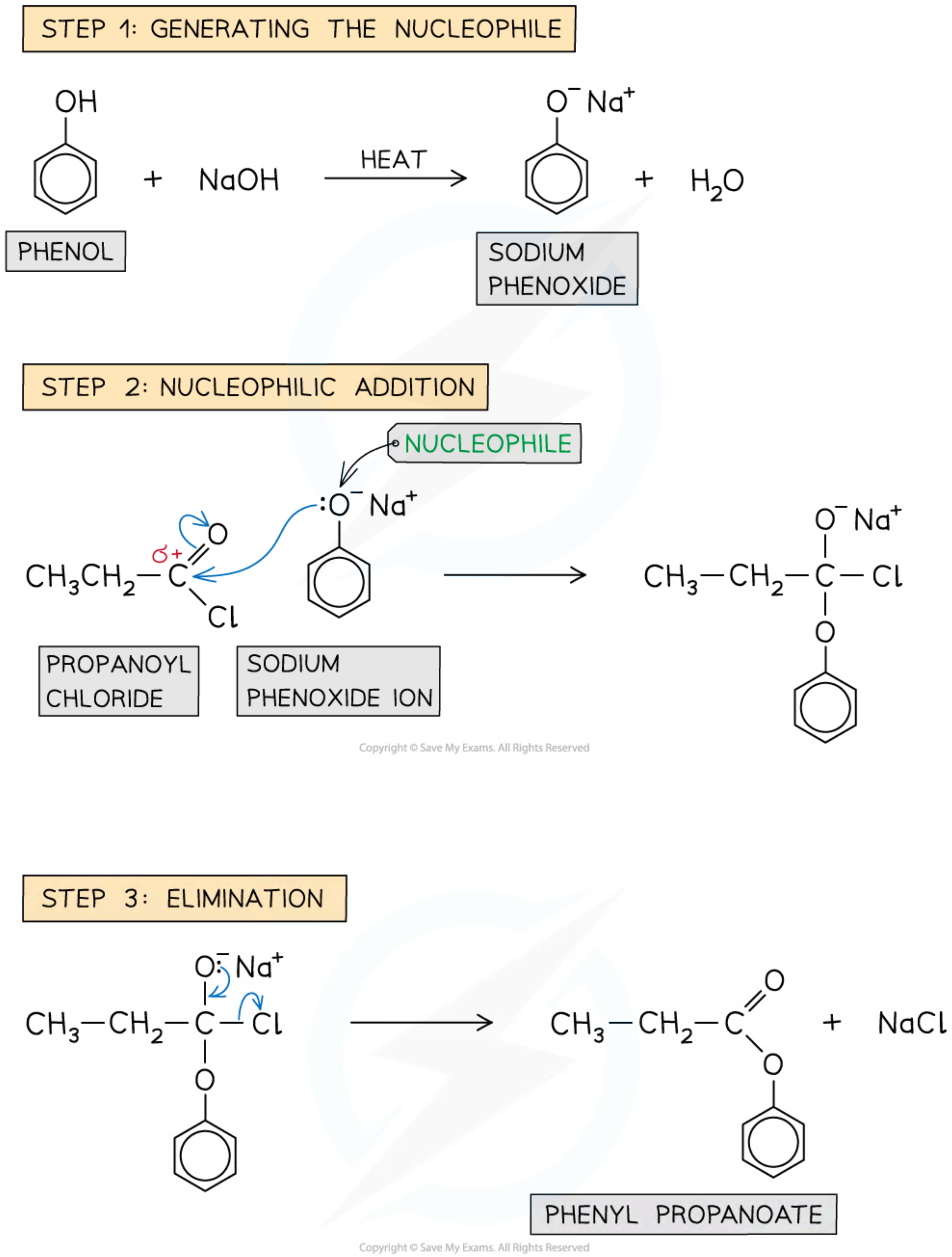
- With phenols, the reaction requires **heat** to proceed and needs to be carried out in the presence of a **base**
- The base **deprotonates** the phenol to form a **phenoxide** ion which is a **better nucleophile** than the phenol molecule
  - The **phenoxide ion** carries out an **initial attack** on the carbonyl carbon
  - A small molecule of NaCl is eliminated

Reaction mechanism of the esterification of acyl chlorides with alcohols



The two-step addition-elimination reaction mechanism of propanoyl chloride and ethanol to form ethyl propanoate and water

Reaction mechanism of the esterification of acyl chlorides with phenols



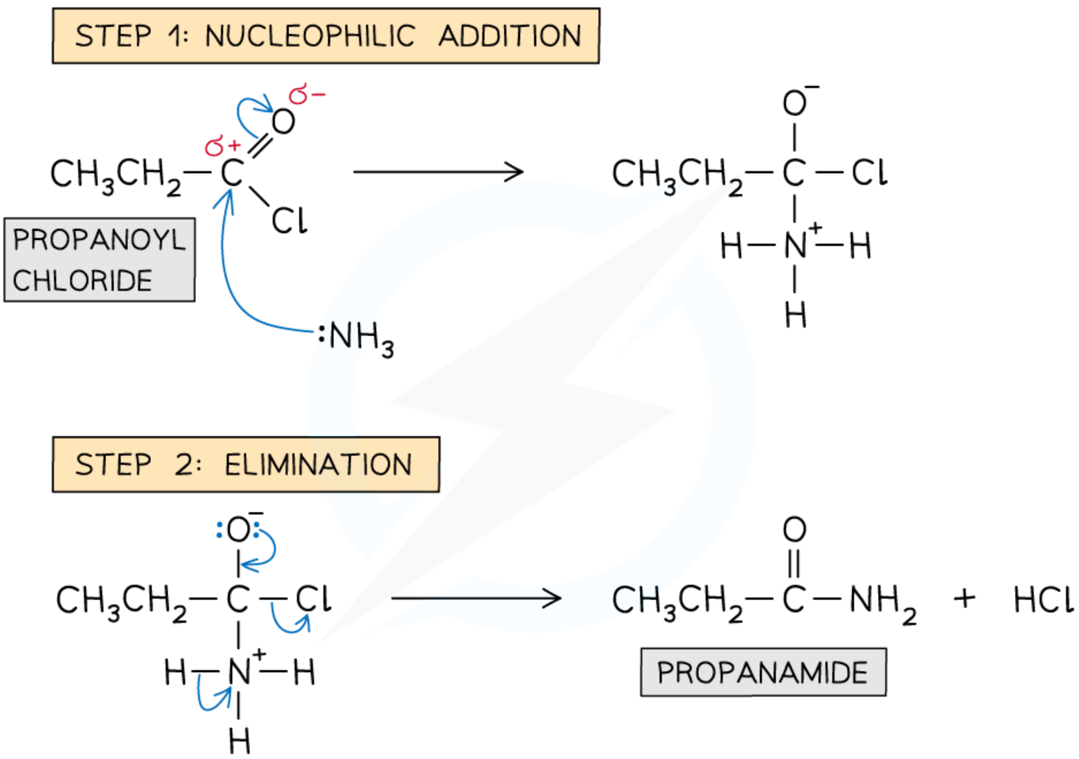
The three-step addition-elimination reaction mechanism of propanoyl chloride with phenol to form phenyl propanoate

Formation of amides: reaction mechanism

- The nitrogen atom in **ammonia** and **primary/secondary amines** act as a **nucleophile**

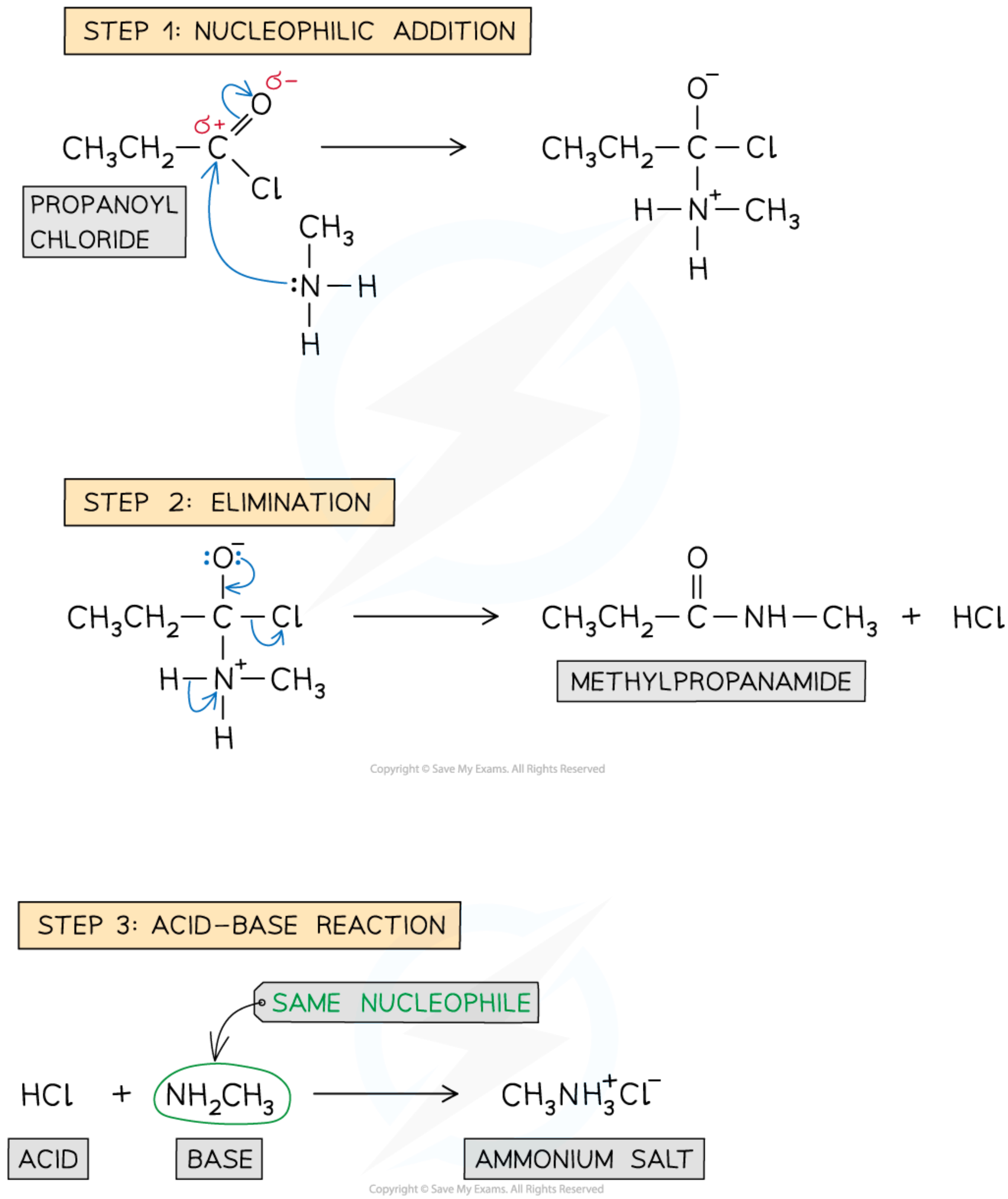
- The lone pair of the nitrogen atom carries out an **initial** attack on the carbonyl carbon
    - This is followed by the elimination of an HCl molecule
- Both reactions of acyl chlorides with ammonia and amines are **vigorous** however there are also differences
    - With **ammonia** - The product is a **non-substituted amide** and **white fumes** of HCl are formed
    - With **amines -** The product is a **substituted amide** and the HCl formed reacts with the **unreacted amine** to form a **white organic ammonium salt**

Reaction mechanism of the formation of amides from acyl chlorides with ammonia



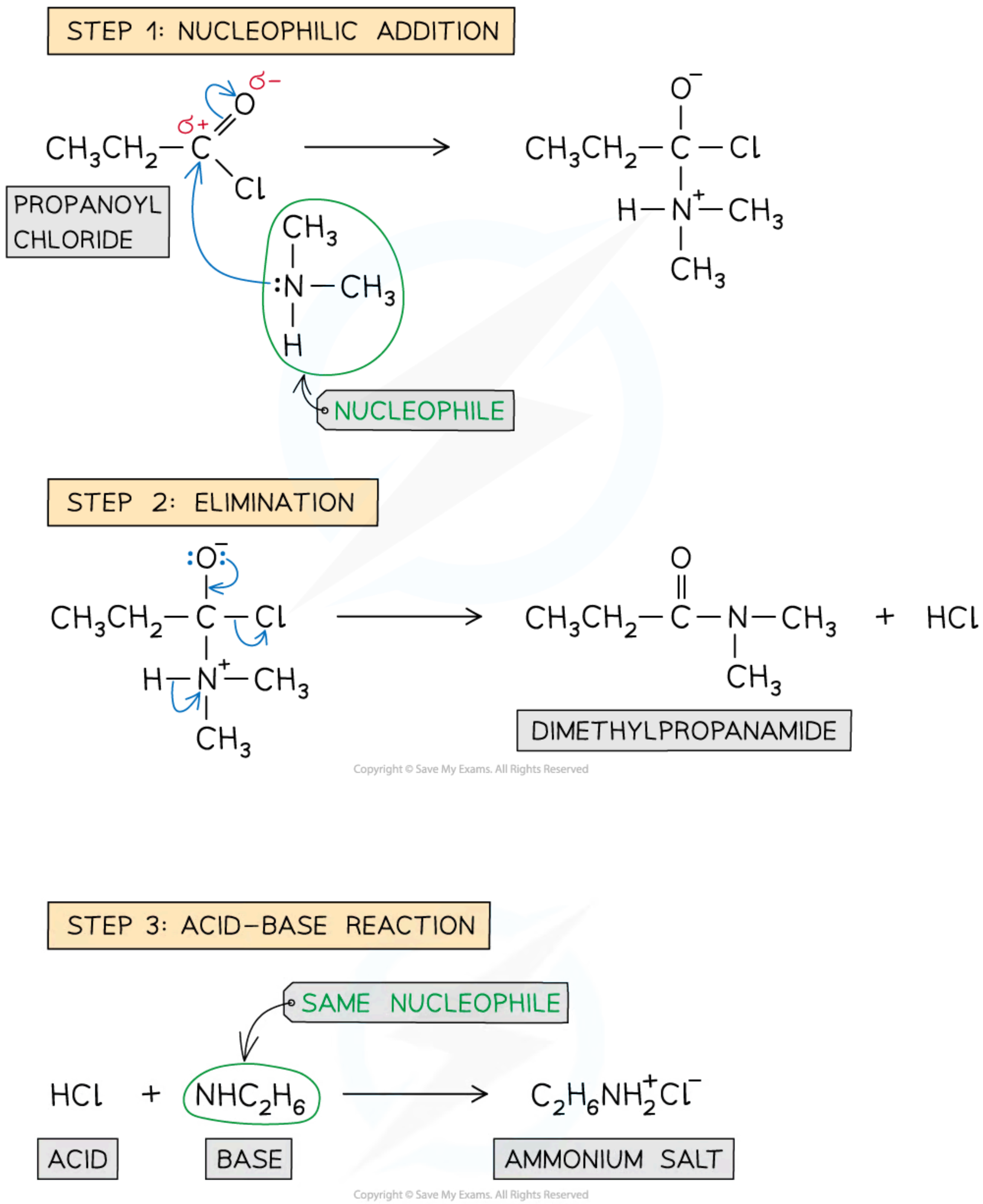
The two-step addition-elimination reaction mechanism of propanoyl chloride and ammonia to form propanamide

Reaction mechanism of the formation of amides from acyl chlorides with primary amines



The addition-elimination reaction mechanism of propanoyl chloride and methylamine to form methylpropanamide

Reaction mechanism of the formation of amides from acyl chlorides with secondary amines



The addition-elimination reaction mechanism of propanoyl chloride and dimethylamine to form dimethylpropanamide